CSCI 232

232 Algorithms & Data Structures

LECTURE: 3

J. L. Pach

OUTLINE:

- Git
  - log
  - branch
  - diff
- Working with Branches

GIT

commands

GIT LS-FILES – LIST TRACKED FILES

This command displays all the files that Git is currently tracking in your repository. It shows the contents of the index (staging area), not the working directory. That means:

- Files listed by git ls-files are already added to Git using git add.
- Untracked files (new files not yet added) will not appear in this list.
- It's useful for checking which files are under version control, especially in large projects.

```
$ git ls-files
firstFile.txt
```

Use git ls-files to verify which files are being tracked by Git. If a file doesn't appear, it's either untracked or ignored via .gitignore.

GIT LOG – COMMIT HISTORY

This command displays the complete commit history of the repository. Each entry includes:

- The commit hash (a unique ID)
- Author name and email
- Date and time of the commit
- The full commit message
- It's useful for reviewing detailed information about each change made to the project.

```
$ git status
On branch master
nothing to commit, working tree clean
```

GIT LOG – EXAMPLE

This command displays the complete commit history of the repository

```
$ git log
commit d06aafaf2cd37f5bc7cd4015656e1ae15241c996 (HEAD -> master)
Author: Jacob Pach <jpach@mtech.edu>
Date:   Thu Aug 28 20:09:17 2025 -0600

    Add firstFile.txt
```

GIT LOG --ONELINE – SIMPLIFIED VIEW

This version shows a condensed list of commits, with:

- A shortened commit hash
- The first line of the commit message

It's ideal for quickly scanning the history or identifying specific commits without all the extra details.

```
$ git log --oneline
d06aafa (HEAD -> master) Add firstFile.txt
```

Use git log when you need full context, and git log --oneline when you want a quick overview. Both are essential tools for navigating and understanding your project's history.

REMOVING COMMITTED FILES IN GIT

Important Note

Be careful when removing files that have already been committed. If, for example, `firstFile.txt` is no longer needed in future commits, and you delete it manually from the repository folder, Git won't automatically recognize this change.

To inform Git that the file was intentionally removed, you must add the deletion using:

```
git add firstFile.txt
```

⚒ This may seem counterintuitive, but it tells Git: *I want this deletion to be part of the next commit.* Once committed, the file will no longer exist in the current branch.

REMOVING COMMITTED FILES IN GIT

Important Note

Of course, you can always restore the file by checking out a previous commit where it still existed.

Git tracks changes — including deletions — only when you explicitly stage them.

Use git add even for removed files to make the change part of your commit history.

RENAMING FILES IN GIT

Important Note

When working with Git, it's important to understand how file renaming is handled. Git does not automatically detect a rename as a single action. Instead, it treats it as:

- Deletion of the old file
- Creation of a new, untracked file

So, if you rename a file manually (e.g., from `oldName.txt` to `newName.txt`), Git will see `oldName.txt` as deleted and `newName.txt` as a new file. To properly reflect this change in Git, you should:

```
git add oldName.txt
git add newName.txt
git commit -m "Renamed file from oldName.txt to newName.txt"
```

SHORTCUT - GIT COMMIT -AM "MESSAGE"

Stage and Commit in a Single Command

The `-am` flag combines staging (**a**dd) and committing (**m**essage) into one step:

```
git commit -am "Your commit message here"
```

- **Saves Time**: Eliminates the separate `git add .` step for modified files.
- **Keeps Focus**: Keeps your workflow fast when making quick, iterative changes.
- **Cleaner Terminal History**: Reduces repetitive status checks and staging commands.

Important Note: This flag only works for modified or deleted tracked files. Brand new (**untracked**) files still require an explicit git add `<file>`.

WHAT IS .GITIGNORE?

The `.gitignore` file tells Git which files or directories to ignore — meaning they won't be tracked, staged, or committed to the repository.

This is useful for:

- Temporary files (e.g., .log, .tmp)
- Build artifacts (e.g., bin/, obj/)
- IDE-specific files (e.g., .vscode/, .DS_Store)
- Secrets or configuration files (e.g., .env, config.local.json)

HOW IT WORKS - .GITIGNORE?

You create a file named `.gitignore` in the root of your repository and list patterns for files or folders you want Git to skip. Example:

```
*.log
*.tmp
node_modules/
.env
```

- Git will ignore any file or folder that matches these patterns — even if they exist in your working directory.
- Use `.gitignore` to keep your repository clean and focused only on the files that matter. It helps avoid accidentally committing sensitive data or unnecessary clutter.

UNDERSTANDING .GITIGNORE – FILE, NOT A FOLDER

It's important to know that `.gitignore` is a **file**, not a folder. The naming convention comes from Linux/Unix systems, where files that start with a dot `.` are treated as hidden or configuration files. From a Windows perspective, `.gitignore` may appear as a file without a name and with an extension, or simply as a special file with no extension, starting with a dot. This can be confusing at first, but it's a common pattern in many development environments.

Examples of similar hidden/config files in Linux:

- `.ssh/` – stores SSH keys and config
- `.vscode/` – stores VS Code workspace settings

These dot-prefixed files and folders are used to configure tools and environments without cluttering the main workspace.

WORKING

with Branches

WHAT IS A GIT BRANCH?

A branch in Git is like a separate line of development. It allows you to work on new features, bug fixes, or experiments without affecting the main codebase. The default branch is usually called main or master.

WHAT IS A GIT BRANCH?

- Branches help teams collaborate safely and efficiently by isolating changes until they're ready to be merged.
- Think of branches as parallel timelines. You can develop safely in one branch, test your changes, and merge them back when everything works. This is a core concept in modern version control.

GIT BRANCH – CREATE A NEW BRANCH

- The command git branch shows a list of all branches in your repository. The currently active branch is marked with an asterisk (*) `git branch new_branch`
- When working with a repository that has multiple branches, we can switch between them using two different commands. This is because modern versions of Git introduced standardized naming conventions, but the older commands were kept to ensure backward compatibility and to avoid forcing experienced users to relearn everything from scratch.

  `$ git switch second_branch` or `$ git checkout second_branch`

GIT STATUS & GIT BRANCH

```
git branch
* master
  second
```

```
git status
On branch master
...
```

Depending on the shell or terminal, Git can display additional information in the prompt, such as the current branch, whether all files are tracked, or if there are uncommitted changes. However, to check which branch you are on, you can always use the git status command or git branch without any parameters.

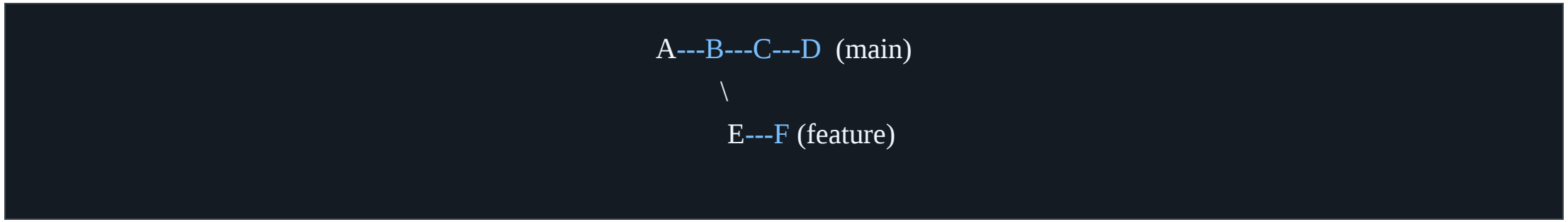
GIT MERGE

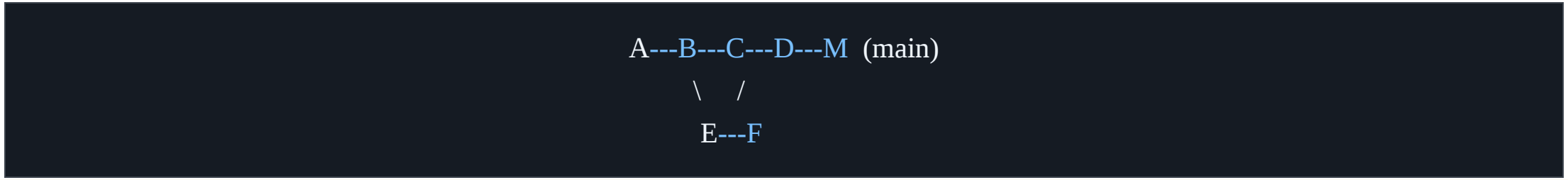
- A merge combines changes from one branch into another.
- **Typically, we merge into the main/master branch.**
- Example workflow:
  - Switch/Checkout main
  - Run git merge feature-branch
  - main now includes the changes from feature-branch

GIT MERGE

A merge creates a merge commit that keeps both branch histories visible. Useful when you want a complete history of how branches diverged and then rejoined.

Run from main





WORKING WITH BRANCHES

Branch inheritance:

- A newly created branch (other than main/master) inherits all files and commits that existed at the moment of its creation.
- In other words: it's a copy of the project's state at that time.

Files created inside a branch:

- New files committed inside a branch "live" only in that branch.
- The main branch (main/master) does not see these files unless a merge happens.

WORKING WITH BRANCHES

`Merge` command:

- The git merge `branch` command merges the given branch into the currently active branch.
- **Important**: the merge is always one-way – from the branch given as a parameter → into the branch you are on.

WHAT HAPPENS TO THE MERGED BRANCH?

- After a merge, the merged branch is not deleted automatically.
- Whether you remove it depends on team strategy and workflow:
  - Deleting keeps the repository clean and avoids clutter.
  - Keeping old branches may obscure the project and cause confusion.
- Remember: deleting a branch is **irreversible** if its changes are not merged.

WORKING WITH BRANCHES

Merge conflicts:

- A merge conflict happens when the same file was modified differently in two branches.
- Conflicts are very common in collaborative projects.
- Expect them – don’t assume merging will always be automatic..

RESOLVING MERGE CONFLICTS

- Git will stop and inform you which files are in conflict.
- Inside the file you'll see conflict markers like:
- The developer must manually decide which parts to keep.
- After editing and saving, run:
- to finalize the merge.

```
<<<<<< HEAD
your current branch content
=======
incoming branch content
>>>>>> feature-branch
```

`git add <file>` and `git commit`

SUMMARY 1/3

Let's review what we have learned so far. We previously worked on creating new branches in Git and switching between them. One important observation is that files seem to be "hidden" when moving between branches, since each branch has its own version of the file system state.

- A newly created branch (other than main or master) inherits everything that existed at the moment it was created.
- Any new files added and committed in this branch "live" only in that branch. The main branch does not automatically have access to them.
- The merge command allows us to bring another branch into the currently active branch (never the other way around!).
- After merging, the absorbed branch is not automatically deleted.

SUMMARY 2/3

It is important to remember that merging cannot be **undone** easily, and the consequences are permanent. On the other hand, in large projects, a huge number of branches may appear to allow independent development of features. This branching helps with safety and organization, but leaving unsupported or abandoned branches can create confusion and reduce clarity in the codebase.

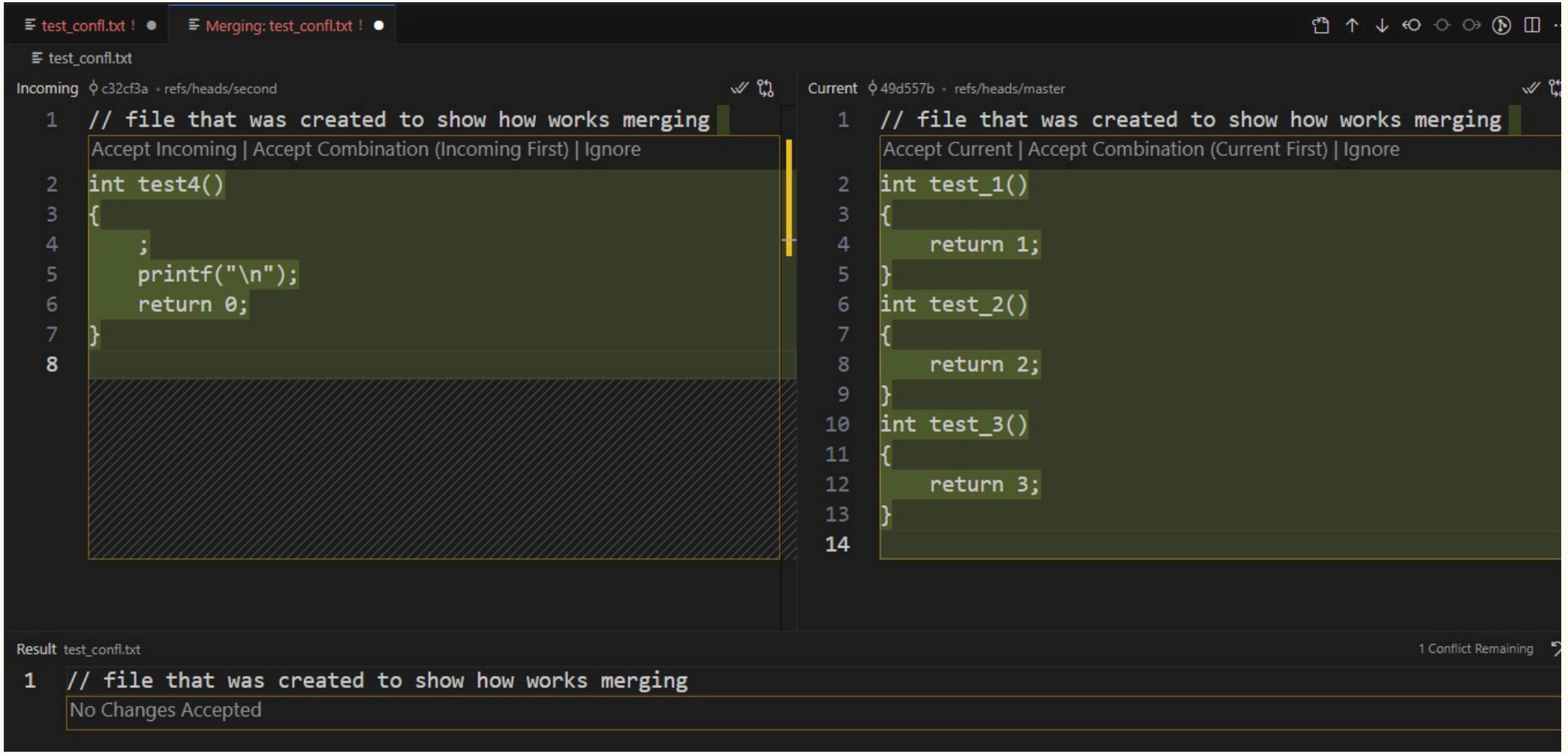
- Merge conflicts are very common. They occur when two branches contain different versions of the same file. Hoping for no conflicts is unrealistic.

SUMMARY 3/3

- When a conflict happens, Git will report it and show which files are affected. Inside the file, Git will mark the conflicting sections with `<<<<<< HEAD`, `======`, and `>>>>>>> branch_name`.
- At this point, the software engineer must manually resolve the conflict: decide which parts of the code remain and which should be removed. After making the necessary edits, the file must be saved, staged (git add), and committed.

In short, branches allow safe and parallel development, merging combines work from different lines of development, and conflicts are a natural part of collaboration that every developer must learn to resolve.

EXAMPLE OF THE MERGING PROCESS



INTRODUCTION TO GIT DIFF

What is git diff?

- git diff is a command used to show changes between commits, branches, or your working directory and the index.
- **It does not change anything** — it only displays differences.
- Important note:
- The +++ and --- lines in the diff output do not indicate whether something was added or removed.
  - They simply represent the two versions being compared:
    - --- → the “old” version
    - +++ → the “new” version

INTRODUCTION TO GIT DIFF

Example:

- Here, the - and + at the start of lines indicate removal or addition. --- and +++ just label the files.

```
diff --git a/file.txt b/file.txt
--- a/file.txt
+++ b/file.txt
@@ -1,3 +1,3 @@
-Hello world
+Hello Git
```

INTRODUCTION TO GIT DIFF

Comparing staged changes: --staged (or --cached)

- By default, git diff compares your working directory with the index (staged area).
- To compare staged changes with the last commit:
- Shows changes that **are staged for the next commit**, i.e., tracked files that have been modified and added with git add.

```
git diff --staged
```

INTRODUCTION TO GIT DIFF

Comparing specific commits:

- You can compare changes between two commits using their commit IDs:
- Replace <commit1> and <commit2> with the **hashes** of the commits you want to compare.
- Example:
- Output shows **what changed from commit 3a5f1b2 to commit 9c7e8d0**.
- Hint: git log --oneline

```
git diff <commit1> <commit2>
```

```
git diff 3a5f1b2 9c7e8d0
```

INTRODUCTION TO GIT DIFF

Order of commit IDs matters:

- git diff A B ≠ git diff B A
- The first commit (A) is considered the “old” version.
- The second commit (B) is considered the “new” version.
- Lines starting with - were removed from A → B, lines starting with + were added in B compared to A.
- Example:

```
git diff A B   # shows changes to get from A to B
git diff B A   # shows changes to get from B to A (reversed)
```

SUMMARY - GIT DIFF

- git diff is a read-only comparison tool.
- +++ / --- do not indicate changes, only file versions.
- Use --staged to check what is staged for commit.
- Use commit IDs to compare specific commits.
- Order matters when comparing commits.
- git diff –staged or git diff --cached

THANK

You